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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name Debug The World

Selected Domain 2

Selected SDG Track (1–17) SDG 9 — Industry, Innovation and Infrastructure

Project / Solution Name ForgeFlow

Name (Leader) ZHOU HONGRUI

Email & Phone (Leader) Phone +60 1117563302 | Email: era2609074@xmu.edu.my

University Affiliation(s) Xiamen University MALASIA

Number of Team Members 5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Many mid-sized manufacturers still keep order information, production updates and quality records in separate spreadsheets, paper documents or chat messages. When customers ask about their orders, sales staff often have to contact managers, who then check with factory workers. This makes even simple updates slow and inefficient. The problem becomes more serious when a product fails inspection. Workers may struggle to find the affected batch, identify the cause and decide what to do next. This affects customers, sales teams, managers and factory workers. Customers face delays and uncertainty, while employees waste time searching for information. For manufacturers, these problems increase operating costs, slow down decision-making and may affect delivery schedules and customer trust.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

ForgeFlow is a browser-based order and product-traceability prototype for mid-sized manufacturers. Customer-side Sales submits an order and follows its status; a Manager reviews the order, records an approval or rejection, and issues the official order number; factory staff check raw materials and finished stock, record production stages and inspections, and update dispatch. Order, production and batch records are linked so the customer-side Sales account and Manager can view the same order status. The dashboard indicates a 15-second refresh interval. In the current prototype, the customer-risk card is a deterministic demo assessment using sample data. It is not a live GenAI model, an external credit lookup, or an automated approval. For Phase 2, we propose a human-in-the-loop AI assistant that can summarize order information and flag missing or unusual entries for Manager review; people retain all approval and release decisions. We plan to apply workshop skills in database design, role-based access, cloud deployment and evaluation. No specific GenAI API or SDC sponsor platform is confirmed for the current prototype. The exact implementation stack and any planned API must be verified against the repository before final submission.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Current demonstrable flow: Sales, Manager and Factory Operator use role-based browser views in one shared ForgeFlow web application hosted on Replit. Orders, inventory, production, batch and inspection records connect through dashboard and traceability views. The dashboard shows a 15-second refresh interval. Manager approvals and factory inspection entries remain human-controlled. The risk card uses deterministic demo logic and sample data; the running prototype does not call a live AI model or external AI API. No sponsor-provided API or cloud resource is confirmed. Planned Phase 2: optionally add a declared GenAI API for order summaries or flags, subject to Manager review.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We mapped the order journey from Sales submission through Manager review, factory inventory and production, inspection, logistics and customer-side status viewing. The prototype uses role-specific views around a shared order and batch history, with stage status, timestamps and inspection outcomes to make the handoffs understandable without specialist training. For Phase 2, we will test common tasks with Sales, Managers and factory operators, then refine navigation based on where users hesitate or need clarification. We will measure internal handoffs per status inquiry, lookup time, inspection record completeness and whether quality holds prevent premature dispatch. The template baseline of two handoffs is a scenario estimate, not measured factory data. Before real deployment, we will verify authentication, permission boundaries, audit history, data isolation, backups and service reliability.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 9 — Industry, Innovation and Infrastructure Target 9.5: Enhance scientific research and upgrade the technological capabilities of industrial sectors. ForgeFlow supports this target by improving digital order management, production visibility and product traceability for mid-sized manufacturers.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In our illustrated workflow, a customer inquiry requires two internal handoffs: from Sales to the Manager, and from the Manager to Factory staff. This is an estimated baseline based on our project scenario, not real factory measurements.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Average number of internal staff handoffs needed to answer one customer order-status inquiry.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	We aim to reduce internal staff handoffs by at least 50%, from two to no more than one per order-status inquiry, through shared digital records. This target will be tested using the Phase 2 prototype.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

Yes No☑

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

Yes No ☑

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

Yes No☑

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] ZHOU HONGRUI

[Date] 23/10/2026